

Thermal Equilibrium - when objects have been in contact and the macroscopic coordinate (variables) have stopped changing.

→ involves an exchange of energy between two objects, or an object & its surroundings

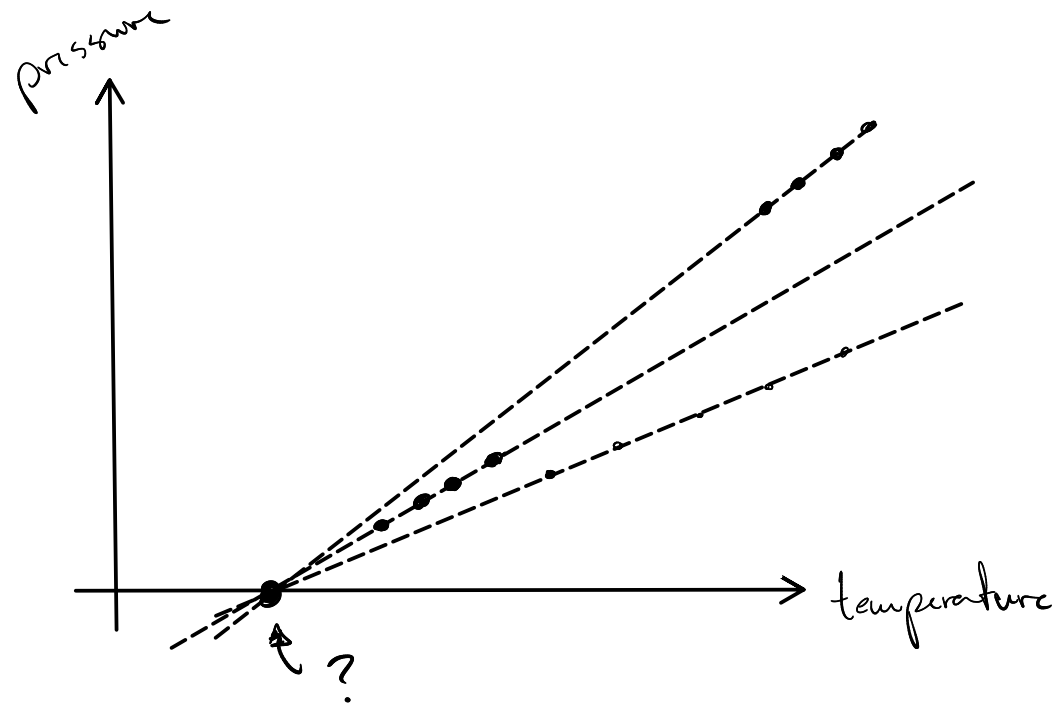
↳ volume

- pressure
- electrical resistance
- radiation / thermal emf

$$\begin{array}{ll} \text{I} & F_1 = 32^\circ\text{F} \quad F_2 = 212^\circ\text{F} \\ & C_1 = 0^\circ\text{C} \quad C_2 = 100^\circ\text{C} \end{array}$$

$$\hookrightarrow C = mF + b$$

$\downarrow$ $\downarrow$
? ?



Equipartition Theorem

$$U_{\text{thermal}} = N \cdot f \cdot \frac{1}{2} k_B T$$

$$f = 3 \leftarrow \text{monatomic gas}$$

$$f = 5 \leftarrow \text{diatomic gas}$$

$$N k_B = n R$$

$$U = n \cdot f \cdot \frac{1}{2} R T$$

$$\hookrightarrow U = \frac{3}{2} n R T \quad (\text{monatomic})$$

$$\hookrightarrow U = \frac{5}{2} n R T \quad (\text{diatomic})$$

Kinetic Energy

$$E_k = \frac{1}{2} m v^2$$

$$\vec{v} = v_x \hat{x} + v_y \hat{y} + v_z \hat{z}$$

$$E_k = \frac{1}{2} m \vec{v} \cdot \vec{v}$$

$$E_k = \frac{1}{2} m (v_x^2 + v_y^2 + v_z^2)$$

Spring Potential

$$U_{sx} = \frac{1}{2} k_x x^2$$

$$U_{sy} = \frac{1}{2} k_y y^2$$

$$U_{sz} = \frac{1}{2} k_z z^2$$

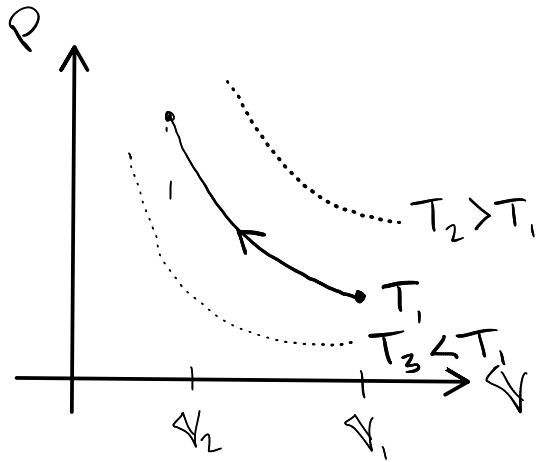
Rotation

$$K_{\text{rot}} = \frac{1}{2} I \omega^2$$

Isobaric (constant pressure) + Isochoric (constant volume), we have discussed in worksheet (week 3)

Isothermal (constant temperature)

↳ compression



$$PV = Nk_B T$$

constant!

$$P = \frac{Nk_B T}{V} \leftrightarrow \text{hyperbola}$$

$$W = - \int P dV$$

$$= -Nk_B T \int_{V_1}^{V_2} \frac{1}{V} dV$$

$$\boxed{W = -Nk_B T \ln\left(\frac{V_2}{V_1}\right)} \rightarrow W = Nk_B T \ln\left(\frac{V_1}{V_2}\right)$$

so what about the heat?

first law $\Delta U = Q + W$

$$\Delta U = \frac{f}{2} Nk_B \Delta T$$

$$Q = -W$$

$$Q = Nk_B T \ln\left(\frac{V_2}{V_1}\right)$$

Adiabatic Process

↳ compression

(isentropic) (no heat added or removed)

first law: $\Delta U = \cancel{Q} + W$

$$\boxed{\frac{f}{2} N k_B \Delta T = W}$$

$$PV = N k_B T$$

$$P = \frac{N k_B T}{V}$$

$$W = \int P dV$$

$$\gamma = - \frac{dy}{dx}$$

↳ $\gamma = \text{~~~~~}$

$$dU = \cancel{dQ} + dW$$

$$dU = dW$$

$$\frac{f}{2} N k_B dT = - P dV$$

$\uparrow$
 $P = \frac{N k_B T}{V}$

$$\frac{f}{2} \cancel{N k_B} dT = - \frac{\cancel{N k_B} T}{V} dV$$

$$\frac{f}{2} dT = - \frac{T dV}{V}$$

$$\frac{f}{2} \int_{T_1}^{T_2} \frac{dT}{T} = - \int_{V_1}^{V_2} \frac{dV}{V}$$

$$\frac{f}{2} \ln\left(\frac{T_2}{T_1}\right) = - \ln\left(\frac{V_2}{V_1}\right)$$

$$\ln\left[\left(\frac{T_2}{T_1}\right)^{\frac{f}{2}}\right] = \ln\left(\frac{V_1}{V_2}\right)$$

$$V \propto \frac{1}{T^{f/2}}$$

$$\left(\frac{T_2}{T_1} \right)^{f/2} = \left(\frac{V_1}{V_2} \right)$$

$$\beta = \frac{\frac{\Delta V}{V}}{\Delta T}$$

$$\beta = \frac{1}{5500 \text{ K}} = 1.81 \cdot 10^{-4} \text{ K}^{-1}$$

$$(a) (0.25 \text{ cm})^3 =$$

$$(2.5 \text{ mm})^3 = \underline{\underline{10 \text{ mm}^3}}$$

$$\beta = \frac{\frac{\Delta V}{V}}{\Delta T}$$

1°C → 1 mm of tube length
1 K

$$1.81 \cdot 10^{-4} \text{ K}^{-1} = \frac{\frac{\Delta V}{10 \text{ mm}^3}}{1 \text{ K}} \Rightarrow \Delta V = 1.81 \cdot 10^{-3}$$

$$\Delta V = \pi r^2 \cdot l$$

$$1.81 \cdot 10^{-3} \text{ mm}^3 = \pi r^2 \cdot 1 \text{ mm}$$

$$r = 0.02 \text{ mm}$$

$$\boxed{d = 0.04 \text{ mm}}$$

8] $\alpha = \frac{\frac{\Delta L}{L}}{\Delta T} \Rightarrow \frac{\Delta L}{L} = \alpha \cdot \Delta T$

does not matter which unit you use $\rightarrow [K^{-1}]$

a] $\alpha = 1.1 \cdot 10^{-5} K^{-1}$

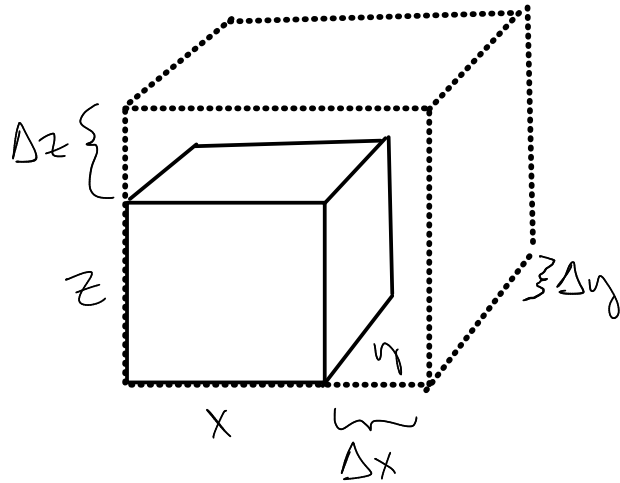
$$\begin{array}{l} T_{\text{cold}} = -10^\circ\text{C} \\ T_{\text{hot}} = 40^\circ\text{C} \end{array} \Rightarrow \Delta T = 50 \text{ K}$$

$$\Delta L = \alpha \cdot \Delta T \cdot L$$

$$\begin{aligned} &= 1.1 \cdot 10^{-5} \cdot 50 \text{ K} \cdot 1 \text{ km} \\ &= 55 \cdot 10^{-5} \text{ km} \cdot \frac{1000 \text{ m}}{1 \text{ km}} = 55 \cdot 10^{-2} = 0.55 \text{ m} = \end{aligned}$$

C

$$\alpha \Delta T = \frac{\Delta L}{L}$$



$$\alpha_x \Delta T = \frac{\Delta x}{x} \quad \left| \quad \alpha_y \Delta T = \frac{\Delta y}{y} \quad \left| \quad \alpha_z \Delta T = \frac{\Delta z}{z} \right. \right.$$

$$\beta \Delta T = \frac{\Delta V}{V} \quad \leftarrow \text{what is } \Delta V \text{ in terms of } \Delta x, \Delta y, \Delta z$$

$$\Delta V = V_f - V_i = (x + \Delta x)(y + \Delta y)(z + \Delta z) - x \cdot y \cdot z$$

$$= (x y + y \Delta x + x \Delta y + \Delta x \Delta y)(z + \Delta z) - x y z$$

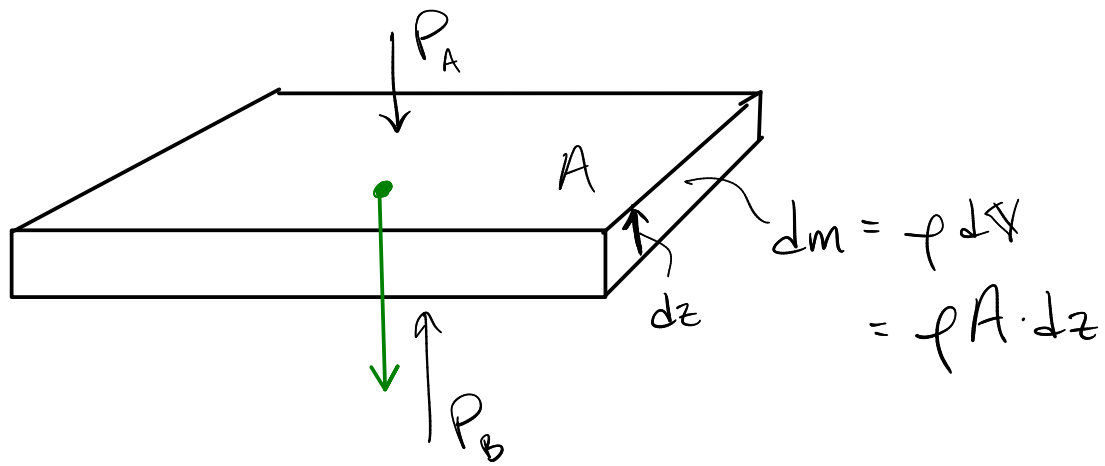
$$= \cancel{x y z} + x y \Delta z + y z \Delta x + y \Delta x \Delta z + x z \Delta y + x \Delta y \Delta z + z \Delta x \Delta y +$$

$$\Delta x \Delta y \Delta z - \cancel{x y z}$$

$$\beta \Delta T = \frac{\Delta V}{V} = \frac{\Delta V}{x y z} = \underbrace{\frac{\Delta z}{z}}_{\alpha_z \Delta T} + \underbrace{\frac{\Delta x}{x}}_{\alpha_x \Delta T} + \cancel{\frac{\Delta x \Delta z}{x z}} + \underbrace{\frac{\Delta y}{y}}_{\alpha_y \Delta T} + \cancel{\frac{\Delta y \Delta z}{y z}} + \cancel{\frac{\Delta x \Delta y}{x y}} + \cancel{\frac{\Delta x \Delta y \Delta z}{x y z}}$$

$$\beta = \alpha_x + \alpha_y + \alpha_z$$

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$$\rho = \frac{M}{V}$$

$$a) \Sigma F_y = \underbrace{P_B \cdot \cancel{A} - P_A \cdot \cancel{A} - \rho \cancel{A} dz \cdot g}_{-dP} = 0$$

$$-dP = \rho g dz$$

$$\boxed{\frac{dP}{dz} = -\rho g}$$

$$b) \frac{dP}{dz} = -\rho g$$

$$= -\frac{mN}{V} \cdot g$$

$$PV = Nk_B T \Rightarrow \frac{N}{V} = \frac{P}{k_B T}$$

$$V = \frac{Nk_B T}{P}$$

$$\frac{dP}{dz} = - \underbrace{\frac{mg}{k_B T}}_C \cdot P$$

$$\frac{dP}{dz} = -C \cdot P$$

$$dP = -C \cdot P \cdot dz$$

$$\int \frac{dP}{P} = \int -C \cdot dz$$

$$\ln P = -Cz + A$$

$$e^{\ln P} = e^{-Cz+A} = e^{-Cz} \cdot e^A = \underbrace{e^A}_A \cdot e^{-Cz}$$

$$P = A e^{-Cz}$$

$$\boxed{P(z) = A e^{-\frac{mgz}{k_B T}}}$$

$$\rightarrow P(0) = A = P_0$$

↑ sea level pressure = 1 atm = $1.013 \cdot 10^5 \text{ Pa}$

$$1.17] \quad a) \quad P = \frac{nRT}{V} \left(1 + \frac{n}{V} B \right)$$

$$P = \frac{n}{V} RT + \left(\frac{n}{V} \right)^2 \cdot BRT$$

$\underbrace{\hspace{1cm}}$
unknown
 x

$$P = x \cdot RT + x^2 \cdot BRT$$

$$0 = BRT \cdot x^2 + RT \cdot x - P$$

$$\frac{n}{V} = x = \frac{-RT \pm \sqrt{(RT)^2 - 4 \cdot BRT(-P)}}{2 \cdot BRT}$$

	T	B	B ²	n/V+	n/V-	n/V ideal	n/V * B
0	100	-160.0	-0.000160	122.747662	6127.252338	120.336943	-0.019640
1	200	-35.0	-0.000035	60.295717	28511.132855	60.168472	-0.002110
2	300	-4.2	-0.000004	40.119075	238055.119021	40.112314	-0.000169
3	400	9.0	0.000009	30.076095	-111141.187206	30.084236	0.000271
4	500	16.9	0.000017	24.057608	-59195.655241	24.067389	0.000407
5	600	21.3	0.000021	20.047597	-46968.404404	20.056157	0.000427

$$\underline{C)} \quad PV = nRT \left(1 + \left(\frac{n}{V} \right) B + \left(\frac{n}{V} \right)^2 C \right) \leftarrow \text{virial}$$

$$\left(P + \frac{an^2}{V^2} \right) (V - nb) = nRT \leftarrow \text{VDW}$$

$$\left(P + \frac{an^2}{V^2} \right) \cdot V \cdot \left(1 - \frac{nb}{V} \right) =$$

$$\left(P + \frac{an^2}{V^2} \right) \cdot V = nRT \left(1 - \frac{nb}{V} \right)^{-1}$$

$$PV + \frac{an^2}{V} =$$

$$PV = nRT \left(1 - \frac{nb}{V} \right)^{-1} - \frac{an^2 \cdot nRT}{V \cdot nRT}$$

$$= nRT \left[\left(1 - \frac{nb}{V} \right)^{-1} - \frac{an}{VRT} \right]$$

$$(1+x)^p \approx 1 + px + \frac{1}{2}p(p-1)x^2$$

$$p = -1 \quad x = -\frac{nb}{V}$$

$$= nRT \left[1 - \left(-\frac{nb}{V}\right) + \frac{1}{2}(-1)(-2)\left(-\frac{nb}{V}\right)^2 - \frac{an}{RTV} \right]$$

$$VDW \rightarrow PV = nRT \left[1 + \frac{nb}{V} + \left(\frac{nb}{V}\right)^2 - \frac{an}{RTV} \right]$$

$\searrow$
 $\frac{n}{V} \left(b - \frac{a}{RT} \right)$

$$\text{virial} \rightarrow PV = nRT \left(1 + \left(\frac{n}{V}\right)B + \left(\frac{n}{V}\right)^2 C \right)$$

$$B = b - \frac{a}{RT} \quad \checkmark$$

$$C = b^2$$

$$C] \quad B = b - \frac{a}{RT}$$

$B(T)$ from data

1.36, 1.40, 1.45

